

Detailed Balance Normalizing Flow for Multi-Modal Boltzmann Sampling

Qi Feng¹, Rongjie Lai², and Xuda Ye²

¹Department of Mathematics, Florida State University, Tallahassee, FL 32306

²Department of Mathematics, Purdue University, West Lafayette, IN 47907

May 4, 2026

1 Introduction

Normalizing flows based on the KL divergence (Rezende and Mohamed, 2015) have been widely applied in generative sampling. In energy-based sampling from physical potentials, the Boltzmann generator (Noé et al., 2019; Wu et al., 2020) has been the industry standard for fast sampling from target Boltzmann distributions. Compared with classical Monte Carlo simulation, which suffers from slow convergence due to long mixing times and strong sample correlation, the Boltzmann generator overcomes this bottleneck and greatly accelerates inference by directly applying the trained flow map to push prior samples to the target, producing independent samples in a single forward pass.

Despite their popularity, KL normalizing flows suffer from several intrinsic drawbacks. First, the flow map is a diffeomorphism on the whole space, so for multi-modal distributions the resulting NF often introduces artificial bridges between modes. Second, the KL loss is highly sensitive to mode collapse, and the flow easily falls into a few dominant modes while neglecting the rest.

In this paper, we propose a normalizing flow based on the detailed balance (DB) divergence, as an alternative to KL-based Boltzmann generators for energy-based sampling from multi-modal Boltzmann distributions. Its key advantage is that the DB divergence matches the gradient rather than the function value and keeps the loss function nonnegative. This incurs only a constant-factor overhead in training while offering substantially more flexibility in use.

2 Detailed balance divergence

2.1 Basic definition

We introduce the detailed balance (DB) divergence to characterize the discrepancy between two distributions. Let μ and ν be probability distributions on $\mathbb{R}^d$. The DB divergence is defined as

$$\text{DB}(\nu\|\mu) = \int_{\mathbb{R}^d} \nu(x) |\nabla \log \nu(x) - \nabla \log \mu(x)| dx = \int_{\mathbb{R}^d} \mu(x) \left| \nabla \left(\frac{\nu(x)}{\mu(x)} \right) \right| dx, \quad (1)$$

which is reminiscent of both the KL divergence

$$\text{KL}(\nu\|\mu) = \int_{\mathbb{R}^d} \nu(x) \log \frac{\nu(x)}{\mu(x)} dx,$$

and the Fisher information

$$\text{Fisher}(\nu\|\mu) = \int_{\mathbb{R}^d} \nu(x) |\nabla \log \nu(x) - \nabla \log \mu(x)|^2 dx.$$

By definition, $\text{DB}(\nu\|\mu)$ is nonsymmetric and nonnegative, and it vanishes precisely when the ratio $\nu(x)/\mu(x)$ is constant, that is, when $\mu = \nu$.

If μ takes the unnormalized form $\mu(x) \propto e^{-U(x)}$ with potential $U(x)$, then

$$\text{DB}(\nu\|\mu) = \int_{\mathbb{R}^d} |\nabla \nu(x) + \nu(x) \nabla U(x)| dx,$$

so minimizing the DB divergence amounts to solving the PDE

$$\nabla \nu(x) + \nu(x) \nabla U(x) = 0, \quad (2)$$

which is exactly the detailed balance condition for sampling $e^{-U(x)}$ via Langevin dynamics. The DB divergence therefore carries a natural physical meaning.

2.2 Relations to other distribution metrics

The DB divergence is closely related to two classical distribution metrics, the Fisher information and the total variation distance, and sits naturally between the two in strength.

Fisher information. The Fisher information of ν relative to μ is

$$\text{Fisher}(\nu\|\mu) = \int_{\mathbb{R}^d} \nu(x) |\nabla \log \nu(x) - \nabla \log \mu(x)|^2 dx,$$

so by (1) the DB divergence is precisely its L^1 counterpart, with the squared L^2 norm of the score difference replaced by an L^1 norm. The Cauchy–Schwarz inequality together with $\int \nu(x) dx = 1$ then yields

$$\text{DB}(\nu\|\mu)^2 \leq \text{Fisher}(\nu\|\mu). \quad (3)$$

Total variation distance. The total variation distance between μ and ν is

$$\text{TV}(\mu, \nu) = \frac{1}{2} \int_{\mathbb{R}^d} |\mu(x) - \nu(x)| dx.$$

Setting $f(x) = \nu(x)/\mu(x) - 1$, so that $\int \mu(x) f(x) dx = 0$ since μ and ν are normalized, we obtain

$$\text{TV}(\mu, \nu) = \frac{1}{2} \int_{\mathbb{R}^d} \mu(x) |f(x)| dx, \quad \text{DB}(\nu\|\mu) = \int_{\mathbb{R}^d} \mu(x) |\nabla f(x)| dx.$$

If μ satisfies a Cheeger inequality (an L^1 Poincaré inequality) with isoperimetric constant $c_P > 0$, applying it to the mean-zero function f yields

$$\text{TV}(\mu, \nu) \leq \frac{1}{2c_P} \text{DB}(\nu\|\mu). \quad (4)$$

Such a Cheeger inequality holds for log-concave distributions and for Gaussian mixtures with sufficient overlap, so the DB divergence controls the total variation distance for these classes.

Combining (3) and (4) yields the hierarchy

$$\text{TV}(\mu, \nu) \leq \frac{1}{2c_P} \text{DB}(\nu\|\mu) \leq \frac{1}{2c_P} \sqrt{\text{Fisher}(\nu\|\mu)}, \quad (5)$$

placing the DB divergence between the total variation distance and the Fisher information. It is weaker than the Fisher information, yet still strong enough to control total variation under regularity assumptions on μ .

To illustrate the hierarchy (5), consider the one-dimensional family

$$\mu_\sigma \propto e^{-U_\sigma(x)}, \quad U_\sigma(x) = \frac{1}{2}x^2 + \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2\sigma^2}}, \quad \nu \propto e^{-V(x)}, \quad V(x) = \frac{1}{2}x^2,$$

indexed by $\sigma \in \{2^0, 2^{-1}, \dots, 2^{-7}\}$. The potentials U_σ and V differ by a Gaussian-shaped bump whose total mass equals σ , so as $\sigma \rightarrow 0$ the perturbation becomes thinner and taller but carries less and less probability mass. Figure 1 shows that the three metrics react to this in very different ways: KL decays at the rate of the bump’s mass and effectively ignores the perturbation; DB treats it as a fixed-size disturbance and stays roughly constant; Fisher amplifies the steep gradient and diverges.

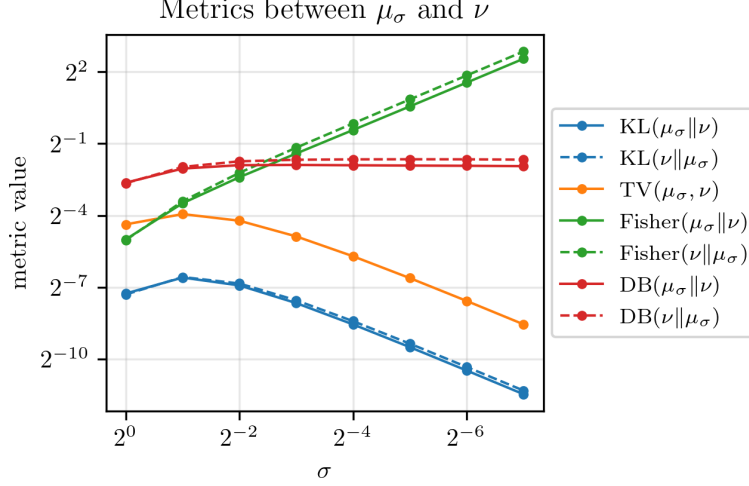


Figure 1: The four metrics between μ_σ and ν for $\sigma = 2^0, 2^{-1}, \dots, 2^{-7}$. KL and TV decay with σ ; DB stays roughly constant; Fisher diverges.

3 Single-step DB normalizing flow

In this section we restrict attention to a single flow map F trained in one shot to push the source onto the target; multi-step compositions are deferred to the next section. Let $U_0(x)$ and $U_1(x)$ be source and target potential functions on $\mathbb{R}^d$, and let $\mu_0(x) \propto \exp(-U_0(x))$ and $\mu_1(x) \propto \exp(-U_1(x))$ be the corresponding Boltzmann distributions. Given explicit samples from μ_0 , we aim to train a single flow map F such that $F_\# \mu_0 = \mu_1$, with pushforward density

$$(F_\# \mu_0)(y) = \frac{\mu_0(x)}{|\det J_F(x)|}, \quad y = F(x), \quad (6)$$

where $J_F(x) \in \mathbb{R}^{d \times d}$ is the Jacobian matrix of F . We construct the loss from the DB divergence between $F_\# \mu_0$ and μ_1 in the reverse direction, $\text{DB}(F_\# \mu_0 \| \mu_1)$, which after a change of variable becomes an integral against the prior and can be estimated without any samples from the target.

3.1 Reverse DB loss and energy-based sampling

First, we define the log importance weight

$$W_F(x) = \log \frac{(F_\# \mu_0)(F(x))}{\mu_1(F(x))} + \text{const} = U_1(F(x)) - U_0(x) - \log |\det J_F(x)|. \quad (7)$$

Viewing $W_F(x)$ as a function of $y = F(x)$, we have

$$\nabla_y W_F(x) = \nabla \log(F_\# \mu_0)(y) - \nabla \log \mu_1(y).$$

Applying the DB divergence in (1) and the change of variable $y = F(x)$,

$$\text{DB}(F_\# \mu_0 \| \mu_1) = \int_{\mathbb{R}^d} (F_\# \mu_0)(y) |\nabla_y W_F(x)| dy = \int_{\mathbb{R}^d} \mu_0(x) |\nabla_y W_F(x)| dx. \quad (8)$$

The gradient $\nabla_y W_F$ admits an explicit chain-rule expansion. Differentiating (7) with respect to $y = F(x)$,

$$\begin{aligned} \nabla_y W_F(x) &= \nabla U_1(F(x)) - \nabla_y (U_0(x) + \log |\det J_F(x)|) \\ &= \nabla U_1(F(x)) - J_F(x)^{-\top} (\nabla U_0(x) + \nabla \log |\det J_F(x)|). \end{aligned} \quad (9)$$

Substituting (9) into (8), the loss in terms of the flow map F takes the form

$$\text{DB}(F_{\#}\mu_0\|\mu_1) = \int_{\mathbb{R}^d} \mu_0(x) \left| \nabla U_1(F(x)) - J_F(x)^{-\top} (\nabla U_0(x) + \nabla \log |\det J_F(x)|) \right| dx. \quad (10)$$

The loss (10) is highly nonlinear in F , primarily because of the explicit appearance of the Jacobian matrix $J_F(x)$ rather than only its determinant. Standard normalizing flow architectures such as RealNVP and NSF support efficient evaluation of the determinant, but assembling the full Jacobian costs at least $O(d^2)$, which becomes prohibitive in high dimensions even with automatic differentiation.

To circumvent this, we bypass the chain-rule expansion (9) altogether and directly approximate the directional derivative $v^\top \nabla_y W_F(x)$ along a random direction $v \in \mathbb{S}^{d-1}$ by central differences in $y = F(x)$. Fix a small step size $\varepsilon > 0$ and define

$$x_{v\pm} = F^{-1}(F(x) \pm \varepsilon v), \quad (11)$$

so that $F(x_{v\pm}) = F(x) \pm \varepsilon v$. Evaluating W_F from (7) at $x_{v\pm}$ and central-differencing in y gives

$$v^\top \nabla_y W_F(x) \approx \frac{W_F(x_{v+}) - W_F(x_{v-})}{2\varepsilon}, \quad (12)$$

which by (7) expands to

$$\begin{aligned} \frac{W_F(x_{v+}) - W_F(x_{v-})}{2\varepsilon} &= \frac{U_1(F(x) + \varepsilon v) - U_1(F(x) - \varepsilon v)}{2\varepsilon} \\ &\quad - \frac{(U_0(x_{v+}) + \log |\det J_F(x_{v+})|) - (U_0(x_{v-}) + \log |\det J_F(x_{v-})|)}{2\varepsilon}. \end{aligned} \quad (13)$$

The right-hand side requires only forward evaluations of U_1 , F^{-1} , U_0 , and $\log |\det J_F|$ — all available cheaply in NSF-style architectures — and never assembles $J_F(x)$ itself. The reverse DB loss (8) is finally estimated by averaging $|v^\top \nabla_y W_F(x)|$ over $x \sim \mu_0$ and $v \sim \text{Unif}(\mathbb{S}^{d-1})$:

$$\mathcal{L}_{\text{reverse}}[F] = \mathbb{E}_{\substack{x \sim \mu_0 \\ v \sim \text{Unif}(\mathbb{S}^{d-1})}} \left| \frac{W_F(x_{v+}) - W_F(x_{v-})}{2\varepsilon} \right|. \quad (14)$$

Remark *The directional-derivative trick is well-matched to the reverse DB loss but not to a Fisher-based variant. For $v \sim \text{Unif}(\mathbb{S}^{d-1})$ and any $g \in \mathbb{R}^d$,*

$$\mathbb{E}_v |v^\top g| = \frac{\Gamma(\frac{d}{2})}{\sqrt{\pi} \Gamma(\frac{d+1}{2})} |g| \sim \sqrt{\frac{2}{\pi d}} |g|, \quad \mathbb{E}_v (v^\top g)^2 = \frac{1}{d} |g|^2,$$

where the asymptotic on the left follows from Stirling's formula. The DB integrand is $|g| = |\nabla_y W_F|$, so $O(\sqrt{d})$ random directions recover the precision of one full-gradient evaluation, costing roughly $\sqrt{d}$ times a single backward pass. The Fisher information involves $|g|^2$, for which $O(d)$ random directions are needed to match a full gradient — comparable to assembling $J_F(x)$ explicitly. The directional-derivative trick therefore makes sense for the DB normalizing flow but offers no asymptotic saving for its Fisher analogue.

3.2 Forward DB loss and mode guiding

Similar to the reverse DB loss, the forward DB loss $\text{DB}(\mu_1\|F_{\#}\mu_0)$ can be written as

$$\text{DB}(\mu_1\|F_{\#}\mu_0) = \int_{\mathbb{R}^d} \mu_1(y) |\nabla_y W_F(x)| dy, \quad (15)$$

the only difference from (8) being that the integration measure is the target μ_1 rather than the prior μ_0 . The same central-difference scheme as in (12) applies: for a fixed $y \in \mathbb{R}^d$, set

$$x_{v\pm} = F^{-1}(y \pm \varepsilon v), \quad (16)$$

so that $F(x_{v\pm}) = y \pm \varepsilon v$, and approximate $v^\top \nabla_y W_F(x)$ at $x = F^{-1}(y)$ by

$$\frac{W_F(x_{v+}) - W_F(x_{v-})}{2\varepsilon} = \frac{U_1(y + \varepsilon v) - U_1(y - \varepsilon v)}{2\varepsilon} - \frac{(U_0(x_{v+}) + \log |\det J_F(x_{v+})|) - (U_0(x_{v-}) + \log |\det J_F(x_{v-})|)}{2\varepsilon}, \quad (17)$$

yielding the forward loss

$$\mathcal{L}_{\text{forward}}[F] = \mathbb{E}_{\substack{y \sim \mu_1 \\ v \sim \text{Unif}(\mathbb{S}^{d-1})}} \left| \frac{W_F(x_{v+}) - W_F(x_{v-})}{2\varepsilon} \right|. \quad (18)$$

The only difference from (14) is that y is drawn from the target μ_1 , so the computation graph tracks y rather than $x = F^{-1}(y)$.

Remark *Unlike the KL divergence, which is invariant under pushforward by a diffeomorphism, the DB divergence is not; in general*

$$\text{DB}(\mu_1 \| F_{\#} \mu_0) \neq \text{DB}(F_{\#}^{-1} \mu_1 \| \mu_0),$$

because a change of variable transforms the score difference by the Jacobian factor $J_F^\top$, whose L^1 norm is not preserved unless F is an isometry. In principle, $\text{DB}(F_{\#}^{-1} \mu_1 \| \mu_0)$ could also serve as a forward DB loss, but we keep $\text{DB}(\mu_1 \| F_{\#} \mu_0)$ because its derivation is simpler.

In general the forward loss $\mathcal{L}_{\text{forward}}[F]$ in (18) cannot be evaluated directly, since it requires explicit samples from the target μ_1 . The crucial difference between the DB and KL losses is that the DB integrand $|\nabla_y W_F(x)|$ is pointwise nonnegative: replacing μ_1 in (18) by an easier-to-sample mode guiding measure $\hat{\mu}_1$ yields a loss $\hat{\mathcal{L}}_{\text{forward}}[F]$ that still vanishes exactly when $F_{\#} \mu_0 \propto \mu_1$ on $\text{supp}(\hat{\mu}_1)$ — the global minimizer is preserved. KL has no such freedom: its integrand $\log(\mu_1 / F_{\#} \mu_0)$ is signed, so any inaccuracy in μ_1 shifts the optimum to $F_{\#} \mu_0 = \hat{\mu}_1$, and KL training therefore demands exact samples from μ_1 . For DB it suffices to take the guiding measure $\hat{\mu}_1$ to be any unnormalized distribution (or simply a positive measure) whose support covers the modes the reverse loss tends to drop. A Gaussian mixture is a convenient form since it can be sampled from

$$\hat{\mu}_1(y) = \sum_{i=1}^n w_i \mathcal{N}(y; x_i, \Sigma_i), \quad w_i \geq 0. \quad (19)$$

Linearity then expands $\hat{\mathcal{L}}_{\text{forward}}[F]$ as a per-component sum,

$$\hat{\mathcal{L}}_{\text{forward}}[F] = \sum_{i=1}^n w_i \mathbb{E}_{\substack{y \sim \mathcal{N}(x_i, \Sigma_i) \\ v \sim \text{Unif}(\mathbb{S}^{d-1})}} \left| \frac{W_F(x_{v+}) - W_F(x_{v-})}{2\varepsilon} \right|, \quad (20)$$

with $x_{v\pm}$ as in (16). The total loss is then simply

$$\mathcal{L}_{\text{total}}[F] = \mathcal{L}_{\text{reverse}}[F] + \hat{\mathcal{L}}_{\text{forward}}[F]. \quad (21)$$

The guidance term supplies mode information from the target side and effectively suppresses the mode collapse that pure energy-based normalizing flows commonly suffer from.

Remark *Mode guidance has also been considered in Jeffreys Flow (Lin et al., 2026), but the construction there is based on the KL divergence and therefore requires accurate samples from the target μ_1 to remain unbiased. The DB-based guidance proposed here is unbiased for any choice of $\hat{\mu}_1$ whose support covers the target modes, which is the property that the signed KL integrand cannot provide.*

For notational convenience, we denote the flow map produced by the DB normalizing flow training as

$$F = \text{DBNF}(U_0, \mu_0; U_1, \hat{\mu}_1), \quad (22)$$

indicating that F minimizes the DB divergence built from the potentials U_0, U_1 , with μ_0 providing the source samples and the guiding measure $\hat{\mu}_1$ providing the auxiliary mode guidance.

3.3 Choice of guiding measure

Although the DB normalizing flow provides a framework which allows to train the flow F from μ_0 to μ_1 with flexible choices guiding measures $\hat{\mu}_1$, without sacrificing the accuracy, the actual choice of the $\hat{\mu}_1$ is delicate.

4 DB Boltzmann generator

References

- Guang Lin, Christian Moya, Di Qi, and Xuda Ye. Jeffreys flow: Robust boltzmann generators for rare event sampling via parallel tempering distillation. *arXiv preprint arXiv:2604.05303*, 2026.
- Frank Noé, Simon Olsson, Jonas Köhler, and Hao Wu. Boltzmann generators: Sampling equilibrium states of many-body systems with deep learning. *Science*, 365(6457):eaaw1147, 2019.
- Danilo Rezende and Shakir Mohamed. Variational inference with normalizing flows. In *International conference on machine learning*, pages 1530–1538. PMLR, 2015.
- Hao Wu, Jonas Köhler, and Frank Noé. Stochastic normalizing flows. *Advances in neural information processing systems*, 33:5933–5944, 2020.